Fixture record fx-0005

Title: Agent-based modelling of evacuation logistics for rural hospitals during hurricanes

Abstract:
An agent-based simulation models patient and staff evacuation logistics for a network of rural hospitals during hurricane landfall, comparing three staged-evacuation protocols under varying road-closure scenarios.

Methods
Study design: Simulation study (agent-based model)
Country: United States
Population: A network of five rural hospitals and their nursing and physician staff
Concept: Agent-based simulation of patient and staff evacuation logistics
Context: Rural hospital hurricane evacuation planning

Results
- A staged evacuation protocol completed evacuation 22% faster than a simultaneous-evacuation protocol in simulation.
- Road-closure scenarios most affected the two hospitals furthest from the regional trauma centre.

Limitations
The authors note the agent behaviours were parameterised from a single prior evacuation exercise and not validated against a real hurricane evacuation.

Funding
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Funder role: Not applicable; the study received no external funding.

Synthetic test data — pipeline conformance fixture, not scientific evidence.